

ESAT PRACTICE TEST SERIES

MATHS 2 + CHEMISTRY

LEVEL 0 • EASY

Maths 2 + Chemistry

Mathematics 1 + Mathematics 2 + Chemistry

Foundation-focused ESAT practice with more direct setup and shorter solution chains.

MODULE

Mathematics 1

MODULE

Mathematics 2

MODULE

Chemistry

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Maths 2 + Chemistry · Level 0 Easy

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Mathematics 2 + Chemistry. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 1 · preserved Mathematics 1 module

Q1 G	Q2 C	Q3 B	Q4 B	Q5 E
Q6 G	Q7 D	Q8 C	Q9 B	Q10 C
Q11 A	Q12 D	Q13 B	Q14 A	Q15 D
Q16 C	Q17 B	Q18 D	Q19 C	Q20 E
Q21 E	Q22 C	Q23 E	Q24 D	Q25 E
Q26 C	Q27 C			

Mathematics 2

Engineering Mock Test 1 · preserved Mathematics 2 module

Q1 A	Q2 F	Q3 D	Q4 A	Q5 D
Q6 D	Q7 F	Q8 B	Q9 D	Q10 D
Q11 B	Q12 E	Q13 E	Q14 B	Q15 C
Q16 A	Q17 D	Q18 A	Q19 C	Q20 B
Q21 A	Q22 A	Q23 C	Q24 C	Q25 D
Q26 C	Q27 D			

Chemistry

Preserved Chemistry Easy source set

Q1 B	Q2 D	Q3 C	Q4 D	Q5 D
Q6 D	Q7 F	Q8 C	Q9 B	Q10 H
Q11 C	Q12 A	Q13 D	Q14 C	Q15 B
Q16 A	Q17 F	Q18 B	Q19 C	Q20 B
Q21 C	Q22 D	Q23 B	Q24 C	Q25 C
Q26 E	Q27 C			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 1 - Mathematics 1

Answer **G****QUESTION 1**

If the 5-digit number $1X2X3$ is divisible by 9, what is X ?

A 0**B** 1**C** 2**D** 3**E** 4**F** 5**G** 6**WORKED SOLUTION**

The digit sum is $1 + X + 2 + X + 3 = 6 + 2X$. For divisibility by 9, $6 + 2X$ must be a multiple of 9. From the choices, $X = 6$ gives 18.

QUESTION 2

If we define $a \star b = a^2 - b^2$, what is $5 \star 4$?

A 1

B 7

C 9

D 16

E 25

F 41

WORKED SOLUTION

Apply the definition directly: $5 \star 4 = 5^2 - 4^2 = 25 - 16 = 9$.

QUESTION 3

Jack tosses two fair dice. What is the probability that the sum of the two values is no less than 9?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number of all possible cases}}.$$

A $\frac{1}{4}$

B $\frac{5}{18}$

C $\frac{1}{3}$

D $\frac{5}{12}$

E $\frac{2}{3}$

WORKED SOLUTION

For two dice, sums at least 9 are 9, 10, 11, 12, giving $4 + 3 + 2 + 1 = 10$ outcomes. So the probability is $10/36 = 5/18$.

M1-04

Mock Test 1 - Mathematics 1

Answer **B**

QUESTION 4

How many 4-digit integers $\overline{abcd}$ satisfy that

- 1 The thousands digit a and the tens digit c are even numbers;
- 2 The hundreds digit b and the unit digit d are odd numbers.
- 3 This number is divisible by 5?

A 80

B 100

C 125

D 250

E 500

WORKED SOLUTION

Since the number is divisible by 5, the final digit must be 5 because d is odd.

Then $a \in \{2, 4, 6, 8\}$, b has 5 choices, and c has 5 choices, giving

$$4 \cdot 5 \cdot 5 \cdot 1 = 100.$$

M1-05

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 5

If $\frac{m}{10} = \frac{6}{n} = \frac{2}{5}$, what is $m + n$?

A 10

B 13

C 14

D 16

E 19

WORKED SOLUTION

From $\frac{m}{10} = \frac{2}{5}$, $m = 4$. From $\frac{6}{n} = \frac{2}{5}$, $30 = 2n$, so $n = 15$. Hence $m + n = 19$.

QUESTION 6

Nancy has 100 marbles. She gives 20% of them to her friend Pedro. Then Nancy gives 25% of what is left to another friend, Ben. Finally, Nancy gives 10% of what is now left in the bag to her brother Jimmy. What percentage of her original bag of marbles does Nancy have left for herself?

A 20

B 30

C $33\frac{1}{3}$

D 38

E 45

F 50

G 54

H 60

WORKED SOLUTION

Track what remains: after 20% is given away, 80 remain; after giving 25% of 80, 60 remain; after giving 10% of 60, 54 remain.

QUESTION 7

What is the sum of all positive 2-digit integers that have a remainder of 1 when divided by 9 and a remainder of 2 when divided by 5?

A 29

B 56

C 84

D 119

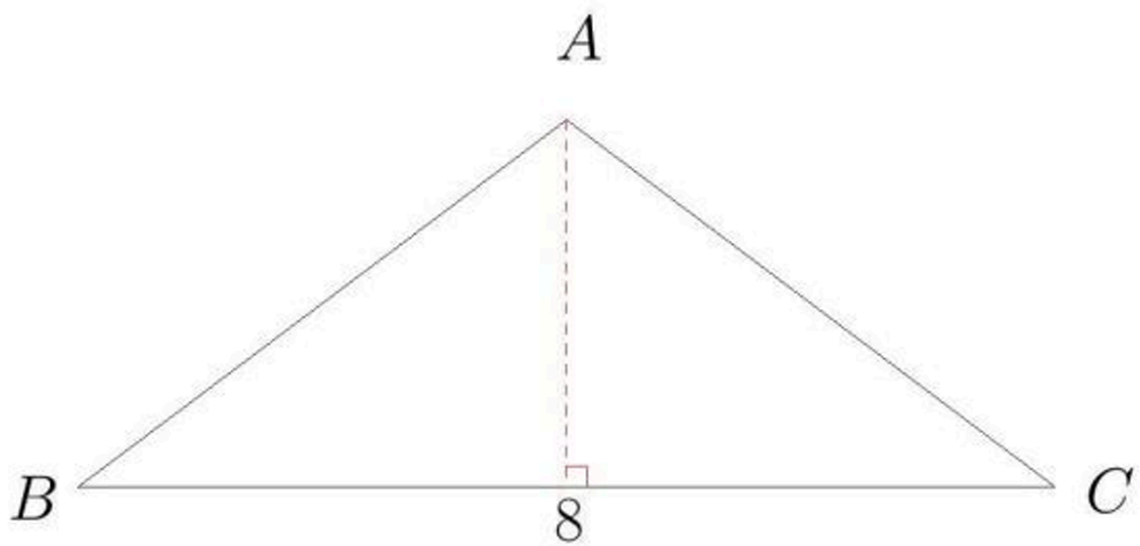
E 164

WORKED SOLUTION

The numbers satisfying both conditions are 37 and 82: each is 1 more than a multiple of 9 and 2 more than a multiple of 5. Their sum is 119.

QUESTION 8

$\triangle ABC$ is an isosceles triangle. $AB = AC$. If $BC = 8$ and the area of $\triangle ABC$ is 12, what is the perimeter of $\triangle ABC$?



A 12

B 16

C 18

D 19

E 20

WORKED SOLUTION

The height is h where $\frac{1}{2} \cdot 8 \cdot h = 12$, so $h = 3$. The altitude bisects the base, giving half-base 4, so each equal side is $\sqrt{3^2 + 4^2} = 5$. Perimeter $= 8 + 5 + 5 = 18$.

QUESTION 9

What is the smaller angle between the hour hand and minute hand of a clock at 5:50 pm?

A 100°

B 125°

C 150°

D 155°

E 165°

WORKED SOLUTION

At 5:50, the minute hand is at 300° . The hour hand is at $5 \cdot 30 + 50 \cdot 0.5 = 175^\circ$. The smaller angle is $300 - 175 = 125^\circ$.

QUESTION 10

The average age of 5 people in a room is 40 years. A 36-year-old person leaves the room. What is the average age of the four remaining people?

A 39

B 40

C 41

D 43

E 45

WORKED SOLUTION

The original total age is $5 \cdot 40 = 200$. After the 36-year-old leaves, the total is 164, so the new average is $164/4 = 41$.

M1-11

Mock Test 1 - Mathematics 1

Answer **A**

QUESTION 11

In how many ways can we write 127 as the sum of two prime numbers?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

An odd number written as a sum of two primes must use 2 plus an odd prime. Here $127 = 2 + 125$, but 125 is not prime, so there are no ways.

QUESTION 12

If for some positive integer n ,

$$\left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{4}\right) \cdots \left(1 + \frac{1}{n}\right) = 50,$$

what is n ?

A 20

B 39

C 65

D 99

E 100

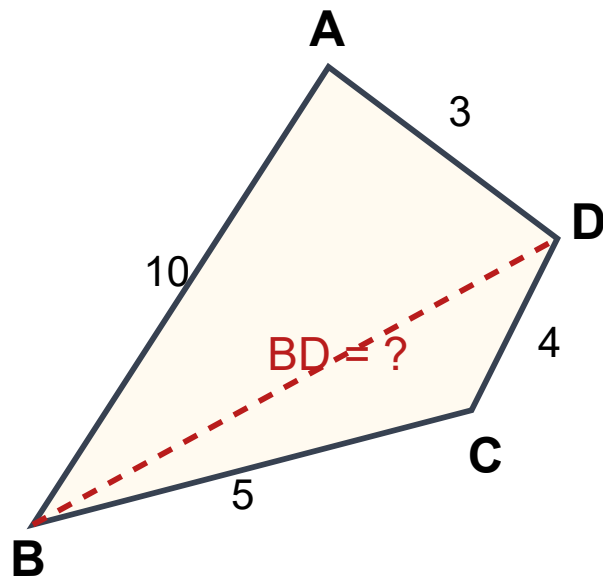
WORKED SOLUTION

Each factor is $1 + \frac{1}{k} = \frac{(k+1)}{k}$. The product telescopes:

$$\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \frac{n+1}{2} = 50. \text{ Thus } n = 99.$$

QUESTION 13

In quadrilateral $ABCD$, $AB = 10$, $BC = 5$, $CD = 4$ and $AD = 3$. If the length of BD is an integer, what is BD ?



A 7

B 8

C 9

D 10

E there is not enough information

WORKED SOLUTION

Use the triangle inequality in the two triangles that share BD . For $\triangle ABD$, $|10 - 3| < BD < 10 + 3$, so $7 < BD < 13$. For $\triangle BCD$, $|5 - 4| < BD < 5 + 4$, so $1 < BD < 9$. The only integer satisfying both is $BD = 8$.

QUESTION 14

What is the unit digit of 7^{2020} ? Here

$$7^n = \underbrace{7 \times 7 \times 7 \times \dots \times 7}_n$$

(Hint: check unit digits of 7 , 7^2 , 7^3 , ... and find the pattern.)

A 1**B** 3**C** 5**D** 7**E** 9**WORKED SOLUTION**

The units digits of powers of 7 cycle as 7, 9, 3, 1. Since 2020 is divisible by 4, the units digit is the fourth term of the cycle, 1.

QUESTION 15

Here x , y , z and u represent four different numbers.

$$\begin{array}{r} x \quad y \quad x \quad y \\ + z \quad x \quad - z \quad x \\ \hline u \quad x \quad \quad x \end{array}$$

$$x + y + z + u =$$

A 12

B 14

C 16

D 18

E 20

WORKED SOLUTION

From the subtraction, $(10x + y) - (10z + x) = x$, so $8x + y = 10z$. From the addition, $(10x + y) + (10z + x) = 10u + x$, so $10x + y + 10z = 10u$. The distinct digit solution is $(x, y, z, u) = (5, 0, 4, 9)$, giving 18.

M1-16

Mock Test 1 - Mathematics 1

Answer C

QUESTION 16

Billy, a seasonal worker in the town of Cowra, collected 8 buckets of cherries on his first day. Each day after that he increased the number of buckets he picked by 2 buckets per day. According to this pattern, how many buckets will he collect in the first 20 days?

A 308

B 500

C 540

D 580

E 600

WORKED SOLUTION

The buckets form an arithmetic sequence: 8, 10, 12, The 20th term is $8 + 19 \cdot 2 = 46$. Sum = $\frac{20}{2}(8 + 46) = 540$.

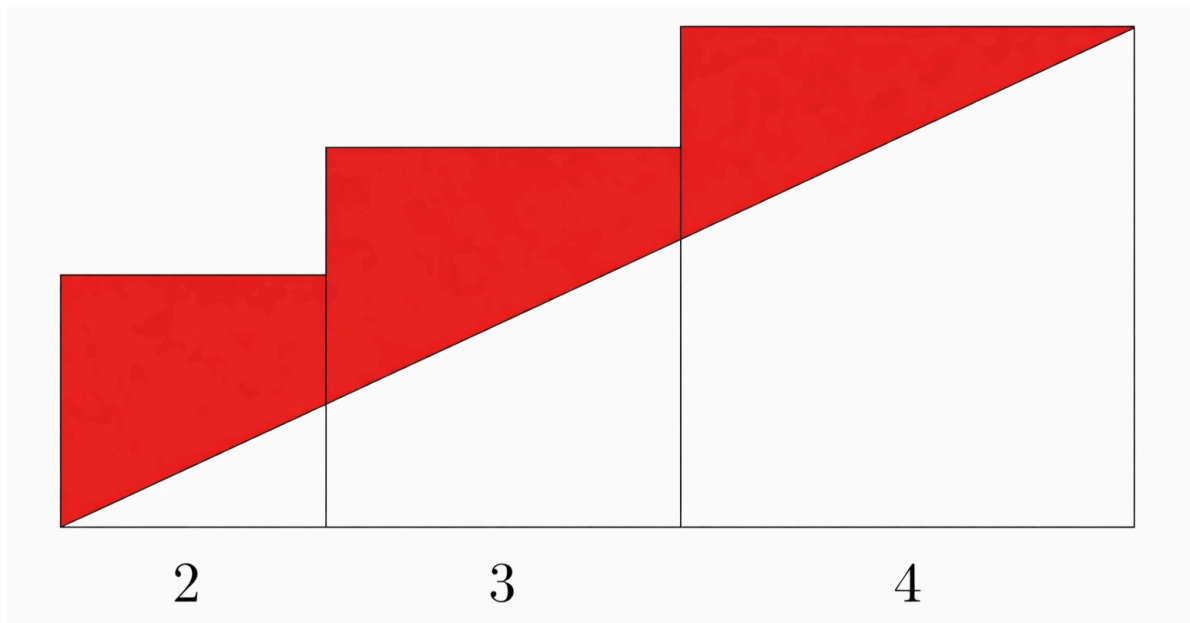
M1-17

Mock Test 1 - Mathematics 1

Answer **B**

QUESTION 17

Three squares are lined up horizontally as shown. Their side lengths are 2, 3 and 4 respectively. A straight line is drawn from the top right corner of the largest square to the bottom left hand corner of the smallest square. What is the area of the shaded region?



A 10

B 11

C 13

D 16

E 18

WORKED SOLUTION

The total area of the three squares is $2^2 + 3^2 + 4^2 = 29$. The line cuts off one large triangle of base 9 and height 4, area 18. Shaded area = $29 - 18 = 11$.

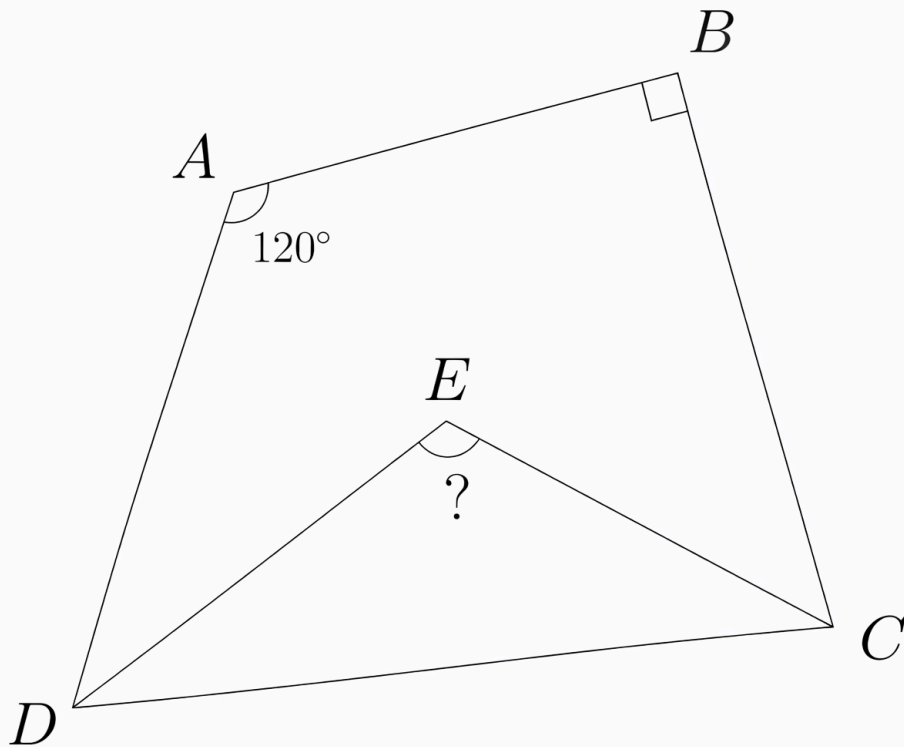
M1-18

Mock Test 1 - Mathematics 1

Answer D

QUESTION 18

In quadrilateral $ABCD$, $\angle A = 120^\circ$ and $\angle B = 90^\circ$. ED and EC bisect $\angle ADC$ and $\angle BCD$ respectively. What is $\angle E$?



A 90

B 95

C 100

D 105

E 110

WORKED SOLUTION

In the quadrilateral, $\angle C + \angle D = 360^\circ - 120^\circ - 90^\circ = 150^\circ$. Since EC and ED bisect those angles, triangle DCE has angles at C, D summing to 75° . Therefore $\angle E = 180^\circ - 75^\circ = 105^\circ$.

M1-19

Mock Test 1 - Mathematics 1

Answer C

QUESTION 19

In a mathematics contest with 10 problems, a student gains 6 points for a correct answer and loses 1 point for an incorrect answer. If Olivia answered every problem and her score was 32, how many correct answers did she have?

A 4

B 5

C 6

D 7

E 8

WORKED SOLUTION

Let c be the number correct. Then incorrect answers are $10 - c$, so
 $6c - (10 - c) = 32$. Hence $7c = 42$ and $c = 6$.

M1-20

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 20

Emily and Owen are going to a library that is 1 mile from their house. They leave home simultaneously. Emily rides her bicycle to the library at a constant speed of 12 miles per hour. Owen walks to the library at a constant speed of 5 miles per hour. How many minutes before Owen does Emily arrive?

A 3

B 4

C 5

D 6

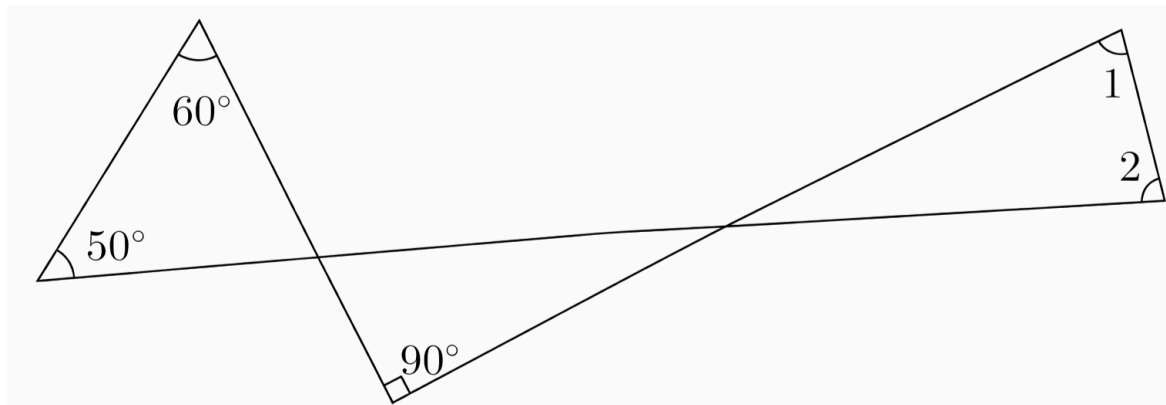
E 7

WORKED SOLUTION

Emily takes $1/12$ hour = 5 minutes. Owen takes $1/5$ hour = 12 minutes. Emily arrives $12 - 5 = 7$ minutes earlier.

QUESTION 21

In the following picture, if $\angle 1 = \angle 2$, what is $\angle 1$?



A 60°

B 65°

C 70°

D 75°

E 80°

WORKED SOLUTION

The left triangle gives the angle at the first intersection as

$180^\circ - 50^\circ - 60^\circ = 70^\circ$. The next triangle has a right angle, so the angle at the second intersection is 20° . The final small triangle has angles x , x , 20° , so $2x = 160^\circ$, giving $x = 80^\circ$.

QUESTION 22

Jack, a zoo keeper, is going to give N peaches to a group of monkeys. If he gives 10 peaches to each monkey, two monkeys will receive no peach. Or, the peaches are exactly enough to give 8 to each monkey. What is the sum of digits of N ?

A 5

B 6

C 8

D 9

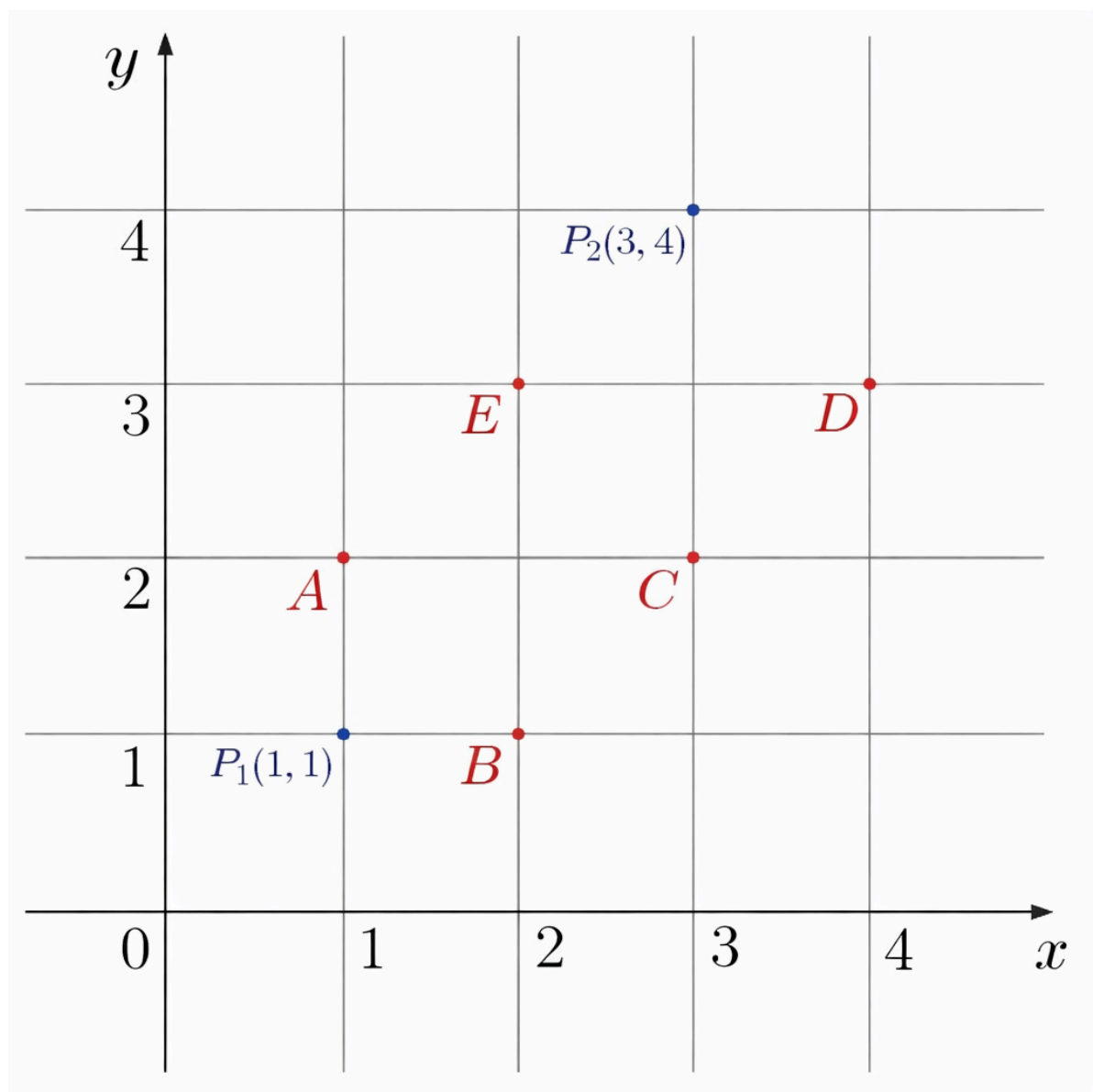
E 10

WORKED SOLUTION

Let there be m monkeys. The two descriptions give $N = 8m$ and $N = 10(m - 2)$. Thus $8m = 10m - 20$, so $m = 10$ and $N = 80$. Digit sum = 8.

QUESTION 23

Every point on the plane can be assigned with a coordinate that has two numbers. For example, the coordinate of point P_1 in the following picture is $(1, 1)$ and the coordinate of point P_2 is $(3, 4)$. Which of the following points has the coordinate $(2, 3)$?



A A

B B

C C

D D

E E

WORKED SOLUTION

Read the grid directly: the point at (2, 3) is labelled E.

M1-24

Mock Test 1 - Mathematics 1

Answer **D**

QUESTION 24

A pizza-shop offers a basic version of pizza with mozzarella and tomatoes. Two different toppings from the following list must be added: anchovies, artichokes, mushrooms, capers. Moreover, for each pizza three different sizes are available: small, medium, large. How many different types of pizza are available at all?

A 11

B 12

C 15

D 18

E 36

WORKED SOLUTION

Choose 2 toppings from 4: $\binom{4}{2} = 6$. For each, there are 3 sizes, so total pizzas $= 6 \cdot 3 = 18$.

M1-25

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 25

Bag A: 1 Snickers bar + 1 Butterfinger bar.

Bag B: 1 Butterfinger bar + 3 Reese's Peanut Butter Cups.

Bag C: 1 Snickers bar + 2 Reese's Peanut Butter Cups.

If each bag costs 50 cents, what is the price of
1 Snickers bar + 1 Butterfinger bar + 1 Reese's Peanut Butter Cup,
in cents?

A 40

B 45

C 50

D 55

E 60

WORKED SOLUTION

Let the prices be S , B , R . The bags give $S + B = 50$, $B + 3R = 50$, and $S + 2R = 50$. Solving gives $S = 30$, $B = 20$, $R = 10$, so $S + B + R = 60$.

M1-26

Mock Test 1 - Mathematics 1

Answer C

QUESTION 26

For how many positive integers x is

$$(x - 1)(x - 4)(x - 9)$$

negative?

A 2

B 3

C 4

D 5

E 6

WORKED SOLUTION

The roots are 1, 4 and 9. Testing the sign in the intervals shows the product is negative for $x < 1$ and $4 < x < 9$. The positive integers satisfying this are 5, 6, 7, 8, so there are 4 values.

M1-27

Mock Test 1 - Mathematics 1

Answer C

QUESTION 27

A gumball machine contains 9 red, 7 white, and 8 blue gumballs. The least number of gumballs a person must buy to be sure of getting four gumballs of the same color is

A 8

B 9

C 10

D 12

E 18

WORKED SOLUTION

The most gumballs possible without having four of one colour is $3 + 3 + 3 = 9$.
One more guarantees four of some colour, so the least number is 10.

Mathematics 2

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Mathematics 2 module. Question wording, option order, answer and solution method are unchanged.

M2-01

Mock Test 1 - Mathematics 2

Answer A

QUESTION 1

Rearrange

$$\frac{7x + 10}{9x + 5} = 3y^2 + 2$$

to make x the subject.

A

$$\frac{15y^2}{7 - 9(3y^2 + 2)}$$

B

$$\frac{15y^2}{7 + 9(3y^2 + 2)}$$

C

$$- \frac{15y^2}{7 - 9(3y^2 + 2)}$$

D

$$- \frac{15y^2}{7 + 9(3y^2 + 2)}$$

E

$$- \frac{5y^2}{7 + 9(3y^2 + 2)}$$

F

$$\frac{5y^2}{7 + 9(3y^2 + 2)}$$

WORKED SOLUTION

Let $A = 3y^2 + 2$. Then $7x + 10 = A(9x + 5)$, so
 $x(7 - 9A) = 5A - 10 = 15y^2$. Therefore $x = \frac{15y^2}{7 - 9(3y^2 + 2)}$.

M2-02

Mock Test 1 - Mathematics 2

Answer **F**

QUESTION 2

Simplify

$$3x \left(\frac{3x^7}{\frac{1}{x^3}} \right)^3$$

A $9x^{20}$

B $27x^{20}$

C $87x^{20}$

D $9x^{21}$

E $27x^{21}$

F $81x^{31}$

WORKED SOLUTION

Dividing by $1/x^3$ is the same as multiplying by x^3 , so the bracket becomes $3x^{10}$.
Cubing gives $27x^{30}$, and multiplying by $3x$ gives $81x^{31}$.

QUESTION 3

For $x \geq 0$, simplify

$$2x[(2x)^7]^{\frac{1}{14}}$$

A $2x\sqrt{2x^4}$

B $2x\sqrt{2x^3}$

C $2\sqrt{2x^4}$

D $2\sqrt{2x^3}$

E $8x^3$

F $8x$

WORKED SOLUTION

$[(2x)^7]^{1/14} = (2x)^{1/2} = \sqrt{2x}$. Then $2x\sqrt{2x} = 2\sqrt{2x^3}$ for $x \geq 0$.

QUESTION 4

What is the circumference of a circle with an area of 10π ?

A $2\pi\sqrt{10}$

B $\pi\sqrt{10}$

C 10π

D 20π

E $\sqrt{10}$

F More information needed.

WORKED SOLUTION

Area = $\pi r^2 = 10\pi$, so $r = \sqrt{10}$. Circumference = $2\pi r = 2\pi\sqrt{10}$.

M2-05

Mock Test 1 - Mathematics 2

Answer **D**

QUESTION 5

If $a \cdot b = (ab) + (a + b)$, then calculate the value of $(3 \cdot 4) \cdot 5$.

A 19

B 54

C 100

D 119

E 132

WORKED SOLUTION

First, $3 \cdot 4 = 3 \cdot 4 + (3 + 4) = 12 + 7 = 19$. Then
 $19 \cdot 5 = 19 \cdot 5 + (19 + 5) = 95 + 24 = 119$.

M2-06

Mock Test 1 - Mathematics 2

Answer **D**

QUESTION 6

If $a \cdot b = \frac{a^b}{a}$, calculate $(2 \cdot 3) \cdot 2$.

A $\frac{16}{3}$

B 1

C 2

D 4

E 8

WORKED SOLUTION

$2 \cdot 3 = 2^3 / 2 = 4$. Then $4 \cdot 2 = 4^2 / 4 = 4$.

M2-07

Mock Test 1 - Mathematics 2

Answer **F**

QUESTION 7

Solve $x^2 + 3x - 5 = 0$.

A $x = -\frac{3}{2} \pm \frac{\sqrt{11}}{2}$

B $x = \frac{3}{2} \pm \frac{\sqrt{11}}{2}$

C $x = -\frac{3}{2} \pm \frac{\sqrt{11}}{4}$

D $x = \frac{3}{2} \pm \frac{\sqrt{11}}{4}$

E $x = \frac{3}{2} \pm \frac{\sqrt{29}}{2}$

F $x = -\frac{3}{2} \pm \frac{\sqrt{29}}{2}$

WORKED SOLUTION

Using the quadratic formula, $x = \frac{-3 \pm \sqrt{3^2 - 4(1)(-5)}}{2} = \frac{-3 \pm \sqrt{29}}{2}$.

M2-08

Mock Test 1 - Mathematics 2

Answer **B**

QUESTION 8

How many times do the curves $y = x^3$ and $y = x^2 + 4x + 14$ intersect?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Intersections satisfy $x^3 = x^2 + 4x + 14$, or $x^3 - x^2 - 4x - 14 = 0$. This cubic has one real root and two complex roots, so the curves intersect once.

M2-09

Mock Test 1 - Mathematics 2

Answer D**QUESTION 9**

Solve for x , y , and z .

1. $x + y - z = -1$
2. $2x - 2y + 3z = 8$
3. $2x - y + 2z = 9$

	x	y	z
A	2	-15	-14
B	15	2	14
C	14	15	-2
D	-2	15	14
E	2	-15	14
F	No solutions possible		

WORKED SOLUTION

Substitute option D: $x = -2, y = 15, z = 14$. Then $-2 + 15 - 14 = -1$, $-4 - 30 + 42 = 8$, and $-4 - 15 + 28 = 9$. It satisfies all three equations.

M2-10

Mock Test 1 - Mathematics 2

Answer **D****QUESTION 10**Fully factorise: $3a^3 - 30a^2 + 75a$

A $3a(a - 3)^3$

B $a(3a - 5)^2$

C $3a(a^2 - 10a + 25)$

D $3a(a - 5)^2$

E $3a(a + 5)^2$

WORKED SOLUTIONFactor out $3a$: $3a^3 - 30a^2 + 75a = 3a(a^2 - 10a + 25) = 3a(a - 5)^2$.**M2-11**

Mock Test 1 - Mathematics 2

Answer **B****QUESTION 11**Solve for x and y :

$4x + 3y = 48$

$3x + 2y = 34$

	x	y
A	8	6
B	6	8

	x	y
C	3	4
D	4	3
E	30	12
F	12	30
G	No solutions possible	

WORKED SOLUTION

Eliminate y : double the first equation gives $8x + 6y = 96$; triple the second gives $9x + 6y = 102$. Subtract to get $x = 6$, then $3(6) + 2y = 34$ gives $y = 8$.

M2-12

Mock Test 1 - Mathematics 2

Answer E

QUESTION 12

Evaluate:

$$- \frac{(5^2 - 4 \times 7)^2}{-6^2 + 2 \times 7}$$

A $-\frac{3}{50}$

B $\frac{11}{22}$

C $-\frac{3}{22}$

D $\frac{9}{50}$

E $\frac{9}{22}$

F 0

WORKED SOLUTION

The numerator is $-(25 - 28)^2 = -9$. The denominator is $-6^2 + 14 = -36 + 14 = -22$. Hence the value is $(-9)/(-22) = 9/22$.

M2-13

Mock Test 1 - Mathematics 2

Answer E

QUESTION 13

All license plates are 6 characters long. The first 3 characters consist of letters and the next 3 characters of numbers. Letters and digits may be repeated. How many unique license plates are possible?

A 676,000

B 6,760,000

C 67,600,000

D 1,757,600

E 17,576,000

F 175,760,000

WORKED SOLUTION

There are 26^3 ways to choose the letters and 10^3 ways to choose the digits. Total $26^3 \cdot 10^3 = 17,576,000$.

M2-14

Mock Test 1 - Mathematics 2

Answer B

QUESTION 14

How many solutions are there for:

$$2(2(x^2 - 3x)) = -9?$$

A 0

B 1

C 2

D 3

E Infinite solutions.

WORKED SOLUTION

The equation becomes $4x^2 - 12x = -9$, or $4x^2 - 12x + 9 = 0$. The discriminant is $(-12)^2 - 4(4)(9) = 0$, so there is exactly one solution.

M2-15

Mock Test 1 - Mathematics 2

Answer C

QUESTION 15

Evaluate:

$$(x^{\frac{1}{2}}y^{-3})^{\frac{1}{2}}$$

A $\frac{\frac{1}{x^2}}{y}$

B $\frac{x}{\frac{3}{y^2}}$

C $\frac{\frac{1}{x^4}}{\frac{3}{y^2}}$

D $\frac{\frac{1}{y^4}}{\frac{3}{x^2}}$

WORKED SOLUTION

Use index laws: $(x^{1/2}y^{-3})^{1/2} = x^{1/4}y^{-3/2} = \frac{x^{1/4}}{y^{3/2}}$.

M2-16

Mock Test 1 - Mathematics 2

Answer A

QUESTION 16

Bryan earned a total of £ 1,240 last week from renting out three flats. From this, he had to pay 10% of the rent from the 1-bedroom flat for repairs, 20% of the rent from the 2-bedroom flat for repairs, and 30% from the 3-bedroom flat for repairs.

The 3-bedroom flat costs twice as much as the 1-bedroom flat. Given that the total repair bill was £ 276, calculate the rent for each apartment.

	1 Bedroom	2 Bedrooms	3 Bedrooms
A	280	400	560
B	140	200	280
C	420	600	840
D	250	300	500

WORKED SOLUTION

Let the 1-bedroom rent be a , the 2-bedroom rent be b , and the 3-bedroom rent be $2a$. Then $3a + b = 1240$, and repairs give $0.1a + 0.2b + 0.3(2a) = 276$.

Solving gives $a = 280$, $b = 400$, and $2a = 560$.

M2-17

Mock Test 1 - Mathematics 2

Answer D

QUESTION 17

At a Pizza Parlour, you can order single, double or triple cheese in the crust. You also have the option to include ham, olives, pepperoni, bell pepper, meat balls, tomato slices, and pineapples. How many different types of pizza are available at the Pizza Parlour?

A 10**B** 96**C** 192**D** 384**E** 768**F** None of the above**WORKED SOLUTION**

There are 3 cheese choices. Each of the 7 toppings can be included or not included, giving 2^7 topping combinations. Total = $3 \cdot 2^7 = 384$.

QUESTION 18

Solve the simultaneous equations $x^2 + y^2 = 1$ and $x + y = \sqrt{2}$, for $x, y > 0$.

A $(x, y) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$

B $(x, y) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$

C $(x, y) = (\sqrt{2} - 1, 1)$

D $(x, y) = \left(\sqrt{2}, \frac{1}{2} \right)$

WORKED SOLUTION

Square $x + y = \sqrt{2}$: $x^2 + y^2 + 2xy = 2$. Since $x^2 + y^2 = 1$, $2xy = 1$. With positive x, y and sum $\sqrt{2}$, this gives $x = y = \sqrt{2}/2$.

QUESTION 19

Which of the following statements is **FALSE**?

A Congruent objects always have the same dimensions and shape.

B Congruent objects can be mirror images of each other.

C Congruent objects do not always have the same angles.

D Congruent objects can be rotations of each other.

E Two triangles are congruent if they have two sides and the included angle of the same magnitude.

WORKED SOLUTION

Congruent shapes have the same side lengths and the same angles. They may be translated, rotated or reflected, but not enlarged or shrunk. Therefore statement C is false.

M2-20

Mock Test 1 - Mathematics 2

Answer **B**

QUESTION 20

Solve the inequality $x^2 \geq 6 - x$.

A $x \leq -3$ and $x \leq 2$

B $x \leq -3$ and $x \geq 2$

C $x \geq -3$ and $x \leq 2$

D $x \geq -3$ and $x \geq 2$

E $x \geq 2$ only

F $x \geq -3$ only

WORKED SOLUTION

Move all terms to one side: $x^2 + x - 6 \geq 0$. Factor: $(x + 3)(x - 2) \geq 0$. The upward-opening quadratic is non-negative outside the roots, so $x \leq -3$ or

$$x \geq 2.$$

M2-21

Mock Test 1 - Mathematics 2

Answer A

QUESTION 21

The hypotenuse of an isosceles right-angled triangle is x cm. What is the area of the triangle in terms of x ?

A $\frac{x^2}{4}$

B $\frac{x}{4}$

C $\frac{3x^2}{4}$

D $\frac{x^2}{10}$

WORKED SOLUTION

Let each equal leg be a . Then $a^2 + a^2 = x^2$, so $a^2 = x^2/2$. Area = $\frac{1}{2}a^2 = \frac{x^2}{4}$.

M2-22

Mock Test 1 - Mathematics 2

Answer A

QUESTION 22

Mr Heard derives a formula:

$$Q = \frac{(X + Y)^2 A}{3B}$$

He doubles the values of X and Y , halves the value of A and triples the value of B . What happens to the value of Q ?

A Decreases by $\frac{1}{3}$

B Increases by $\frac{1}{3}$

C Decreases by $\frac{2}{3}$

D Increases by $\frac{2}{3}$

E Increases by $\frac{4}{3}$

F Decreases by $\frac{4}{3}$

WORKED SOLUTION

Doubling both X and Y makes $(X + Y)^2$ four times larger. Halving A makes the factor 2, and tripling B divides by 3. New $Q = (2/3)$ of old Q , so it decreases by $1/3$.

M2-23

Mock Test 1 - Mathematics 2

Answer C

QUESTION 23

Consider the graphs $y = x^2 - 2x + 3$, and $y = x^2 - 6x - 10$. Which of the following is true?

A Both equations intersect the x-axis.

B Neither equation intersects the x-axis.

C The first equation does not intersect the x-axis; the second equation intersects the x-axis.

- D** The first equation intersects the x-axis; the second equation does not intersect the x-axis.

WORKED SOLUTION

Use the discriminant. For $x^2 - 2x + 3$, $b^2 - 4ac = 4 - 12 = -8 < 0$, so no x-intercepts. For $x^2 - 6x - 10$, $36 + 40 = 76 > 0$, so it intersects the x-axis.

M2-24

Mock Test 1 - Mathematics 2

Answer C

QUESTION 24

Evaluate

$$\log_2\left(\frac{5}{4}\right) + \log_2\left(\frac{6}{5}\right) + \log_2\left(\frac{7}{6}\right) + \cdots + \log_2\left(\frac{64}{63}\right)$$

A -2

B 3

C 4

D 6

WORKED SOLUTION

The logarithms telescope:

$$\log_2(5/4) + \log_2(6/5) + \cdots + \log_2(64/63) = \log_2(64/4) = \log_2(16) = 4.$$

QUESTION 25

Given that $7 \cos \theta - 3 \tan \theta \sin \theta = 1$, which one of the following is true?

A $\cos \theta = -\frac{3}{5}$ or $\cos \theta = -\frac{1}{2}$

B $\cos \theta = -\frac{3}{5}$ or $\cos \theta = \frac{1}{2}$

C $\cos \theta = \frac{3}{5}$ or $\cos \theta = \frac{1}{2}$

D $\cos \theta = \frac{3}{5}$ or $\cos \theta = -\frac{1}{2}$

WORKED SOLUTION

Let $c = \cos \theta$. Since $\tan \theta \sin \theta = \frac{\sin^2 \theta}{\cos \theta} = \frac{(1 - c^2)}{c}$, multiply the equation by c : $7c^2 - 3(1 - c^2) = c$. Thus

$10c^2 - c - 3 = 0 = (5c - 3)(2c + 1)$, so $c = \frac{3}{5}$ or $c = -\frac{1}{2}$.

QUESTION 26

k is the smallest positive value of x which is a solution to both the equations $2 \sin x + 1 = 0$ and $2 \cos 2x = 1$.

How many values of x in the range $0 \leq x \leq k$ are solutions to at least one of these equations?

A 0

B 2

C 3

D 4

E 8

WORKED SOLUTION

$2 \sin x + 1 = 0$ gives $\sin x = -1/2$. The first positive common solution with $2 \cos 2x = 1$ is $k = 7\pi/6$. In $[0, k]$, the solutions to at least one equation are $\pi/6$, $5\pi/6$, and $7\pi/6$: three values.

M2-27

Mock Test 1 - Mathematics 2

Answer D

QUESTION 27

How many solutions of the equation $2 \sin^3 \theta = \sin \theta$ lie in the interval $-\frac{\pi}{2} \leq \theta \leq \pi$?

A 2

B 3

C 4

D 5

E 6

F 7

WORKED SOLUTION

Factor: $2 \sin^3 \theta = \sin \theta \Rightarrow \sin \theta (2 \sin^2 \theta - 1) = 0$. In the interval, $\sin \theta = 0$ gives $0, \pi$; $\sin \theta = \sqrt{2}/2$ gives $\pi/4, 3\pi/4$; $\sin \theta = -\sqrt{2}/2$ gives $-\pi/4$. Total 5 solutions.

Chemistry

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Chemistry Easy source set. Question wording, option order, answer and solution method are unchanged.

C-01

Periodic Table and electron configuration

Answer B**QUESTION 1**

Four atoms have the following electron configurations.

atom	electron configuration
P	2,8,8
Q	2,8,1
R	2,8,7
S	2,8,8,1

Which statement is correct?

- A** P is a Period 2 noble gas.
- B** Q and S are in Group 1, and S is expected to react more vigorously with water than Q.
- C** R normally forms R^{2-} ions.
- D** P has one electron in its outer shell.
- E** Q is less metallic than R.

WORKED SOLUTION

Q has electron configuration 2,8,1 and S has electron configuration 2,8,8,1, so both are in Group 1.

Group 1 reactivity increases down the group. S has one more occupied electron shell than Q, so S is below Q and is expected to react more vigorously with water.

P has three occupied shells and a full outer shell, so it is a Period 3 noble gas. R has seven outer-shell electrons and normally forms a 1^- ion.

Therefore the correct answer is **B**.

C-02

Formulae of ionic compounds

Answer **D**

QUESTION 2

Aluminium ions are Al^{3+} and sulfate ions are SO_4^{2-} .

Which row gives the formula of aluminium sulfate and the number of sulfate ions in one formula unit?

option	formula	sulfate ions
A	AlSO_4	1
B	$\text{Al}(\text{SO}_4)_2$	2
C	Al_2SO_4	1
D	$\text{Al}_2(\text{SO}_4)_3$	3
E	$\text{Al}_3(\text{SO}_4)_2$	2

A A

B B

C C

D D

E E

WORKED SOLUTION

The smallest common magnitude of charge is 6:

$$2 \times (+3) = +6, \quad 3 \times (-2) = -6.$$

Therefore two aluminium ions combine with three sulfate ions:



There are three sulfate ions in one formula unit, so the correct row is **D**.

C-03

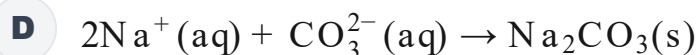
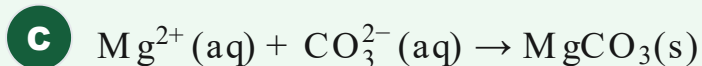
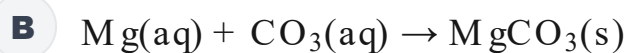
Ionic equations

Answer C

QUESTION 3

Aqueous magnesium chloride is mixed with aqueous sodium carbonate. A white precipitate of magnesium carbonate forms.

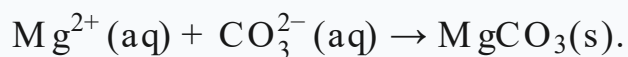
Which is the ionic equation for this reaction?



WORKED SOLUTION

The sodium and chloride ions remain dissolved and are spectator ions.

The ions that actually form the precipitate are magnesium ions and carbonate ions:



Therefore the answer is **C**.

C-04

Moles and particles

Answer **D**

QUESTION 4

How many moles of helium atoms contain the same total number of **atoms** as 0.20 mol of oxygen molecules, O_2 ?

A 0.10 mol

B 0.20 mol

C 0.25 mol

D 0.40 mol

E 0.80 mol

WORKED SOLUTION

Each oxygen molecule contains two oxygen atoms. Therefore 0.20 mol of O_2 contains

$$2(0.20) = 0.40 \text{ mol}$$

of atoms.

Helium is monatomic, so 0.40 mol of helium contains the same number of atoms.

The answer is **D**.

QUESTION 5

A 2.50 g impure sample of calcium carbonate contains impurities that do not react with hydrochloric acid.

The sample is reacted with excess hydrochloric acid and produces 0.480 dm^3 of carbon dioxide at room temperature and pressure.

At rtp,

$$1 \text{ mol of gas} = 24.0 \text{ dm}^3.$$

$$A_r : \quad \text{Ca} = 40, \text{C} = 12, \text{O} = 16$$

What is the percentage purity of the calcium carbonate?

A 40%

B 60%

C 75%

D 80%

E 90%

WORKED SOLUTION

The amount of carbon dioxide formed is

$$n(\text{CO}_2) = \frac{0.480}{24.0} = 0.0200 \text{ mol.}$$

From



the sample contained 0.0200 mol of calcium carbonate.

$$M_r(\text{CaCO}_3) = 40 + 12 + 3(16) = 100.$$

So the mass of pure calcium carbonate was

$$0.0200 \times 100 = 2.00 \text{ g.}$$

Hence

$$\text{purity} = \frac{2.00}{2.50} \times 100 = 80\%.$$

The answer is **D**.

C-06

Kinetic particle theory

Answer **D**

QUESTION 6

A pure substance is heated. During melting, energy continues to be supplied but its temperature remains constant until all of the solid has melted.

Which statement best explains this?

- A** The particles stop moving while the substance melts.
- B** The supplied energy breaks the atoms into smaller particles.
- C** The supplied energy increases the average kinetic energy of the particles, but the temperature does not change.
- D** The supplied energy is used in overcoming forces between particles rather than increasing their average kinetic energy.
- E** Particles in the solid and liquid have exactly the same arrangement.

WORKED SOLUTION

Temperature is related to the average kinetic energy of the particles.

During melting the temperature is constant, so the average kinetic energy is not increasing. The supplied energy is instead used to overcome forces holding particles in the solid structure.

Therefore the correct answer is **D**.

C-07

Metallic bonding

Answer **F**

QUESTION 7

Consider the following statements about a solid metal.

1. Some electrons are delocalised and can move through the structure.
2. Electrical conduction occurs because positive metal ions move through the solid.
3. Layers of metal ions can move relative to one another while metallic bonding is maintained.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: metals contain delocalised electrons that can move through the structure.

Statement 2 is false: the positive metal ions are fixed in the solid lattice; electrical conduction is due to mobile delocalised electrons.

Statement 3 is correct: layers can slide while attraction between positive ions and delocalised electrons is maintained.

So statements 1 and 3 only are correct. The answer is **F**.

C-08

Giant structures

Answer C

QUESTION 8

Which substance has a very high melting point, does not conduct electricity when solid, and does not become an electrical conductor simply on melting?

A sodium chloride

B graphite

C silicon dioxide

D copper

E methane

WORKED SOLUTION

Silicon dioxide has a giant covalent structure. Many strong covalent bonds must be overcome to melt it, so its melting point is very high.

It does not have mobile ions or delocalised electrons available for electrical conduction, whether solid or molten.

Sodium chloride is different because its ions become mobile when molten.

Therefore the answer is **C**.

C-09

Group 1 trends

Answer **B**

QUESTION 9

Which row correctly shows the trends from lithium to sodium to potassium?

option	reactivity with water	melting point
A	$\text{Li} > \text{Na} > \text{K}$	$\text{Li} > \text{Na} > \text{K}$
B	$\text{K} > \text{Na} > \text{Li}$	$\text{Li} > \text{Na} > \text{K}$
C	$\text{K} > \text{Na} > \text{Li}$	$\text{K} > \text{Na} > \text{Li}$
D	$\text{Li} > \text{Na} > \text{K}$	$\text{K} > \text{Na} > \text{Li}$
E	$\text{Na} > \text{K} > \text{Li}$	$\text{Li} > \text{K} > \text{Na}$

A A

B B

C C

D D

E E

WORKED SOLUTION

Group 1 metals become more reactive down the group:



Their melting points decrease down the group:



Only row **B** gives both trends correctly.

C-10

Air and drinking water

Answer **H**

QUESTION 10

Consider these statements.

1. Dry air contains approximately 78% nitrogen.
2. Chlorine may be added during drinking-water treatment to kill harmful microorganisms.
3. Fluoride ions may be added to drinking water to help reduce tooth decay.
4. Components of air can be separated by fractional distillation after the air has been liquefied.

Which statements are correct?

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 4 only
- D** 1, 2 and 3 only
- E** 1, 2 and 4 only
- F** 1, 3 and 4 only
- G** 2, 3 and 4 only
- H** 1, 2, 3 and 4

WORKED SOLUTION

All four statements are correct.

Dry air is about 78% nitrogen. Chlorine is used for disinfection in water treatment, fluoride ions may be added to help reduce tooth decay, and the major gases in air can be separated by fractional distillation after liquefaction.

Therefore the answer is **H**.

C-11

Separation techniques

Answer **C**

QUESTION 11

A mixture contains:

- insoluble sand,
- hexane,
- water containing dissolved sodium chloride.

Hexane and water are immiscible.

Which sequence is suitable for first removing the sand, then isolating the hexane, and finally obtaining sodium chloride crystals?

A separating funnel → filtration → evaporation

B filtration → crystallisation → separating funnel

C filtration → separating funnel → crystallisation

D crystallisation → filtration → separating funnel

E centrifugation → chromatography → filtration

WORKED SOLUTION

First remove the insoluble sand by filtration.

The filtrate contains two immiscible liquid layers, hexane and aqueous sodium chloride, so a separating funnel can isolate the hexane.

Finally, sodium chloride crystals can be obtained from the aqueous layer by crystallisation.

The correct sequence is therefore **filtration** → **separating funnel** → **crystallisation**, answer **C**.

C-12

Obtaining a pure solvent

Answer **A**

QUESTION 12

A mixture contains water, dissolved sodium chloride, and a small amount of insoluble charcoal.

Which procedure is most suitable for obtaining a sample of **pure water** from the mixture?

- A** Filter, then use simple distillation on the filtrate.
- B** Filter, then evaporate the filtrate to dryness.
- C** Use a separating funnel, then crystallise.
- D** Use fractional distillation without first filtering.
- E** Use paper chromatography.

WORKED SOLUTION

The insoluble charcoal should first be removed by filtration.

The filtrate is salt water. Sodium chloride is non-volatile, while water can be vapourised and condensed, so simple distillation gives pure water as the distillate.

Therefore the answer is **A**.

QUESTION 13

A strong acid has pH 2.

It is diluted with water until the total volume is 100 times its original volume. Assume the acid remains fully ionised.

What is its new pH?

A 1

B 2

C 3

D 4

E 6

WORKED SOLUTION

A 100-fold dilution reduces the hydrogen-ion concentration by a factor of

$$100 = 10^2.$$

Each factor of 10 reduction in $[H^+]$ increases pH by 1, so the pH increases by 2:

$$2 + 2 = 4.$$

Therefore the answer is **D**.

QUESTION 14

20.0 cm³ of 0.30 mol dm⁻³ sulfuric acid is completely neutralised by 0.50 mol dm⁻³ sodium hydroxide.

Assume one mole of sulfuric acid supplies two moles of H⁺.

What volume of sodium hydroxide is required?

A 12.0 cm³

B 18.0 cm³

C 24.0 cm³

D 30.0 cm³

E 48.0 cm³

WORKED SOLUTION

The amount of sulfuric acid is

$$n(\text{H}_2\text{SO}_4) = 0.30 \times 0.0200 = 0.00600 \text{ mol.}$$

Each mole supplies two moles of H⁺, so 0.0120 mol of OH⁻ is needed.

Therefore

$$V(\text{NaOH}) = \frac{0.0120}{0.50} = 0.0240 \text{ dm}^3 = 24.0 \text{ cm}^3.$$

The answer is **C**.

QUESTION 15

A student wants to prepare pure, dry crystals of copper(II) sulfate using dilute sulfuric acid and insoluble copper(II) oxide.

Which method is best?

- A** Add exactly one spatula of copper oxide to cold acid and evaporate immediately to dryness.
- B** Warm the acid, add copper oxide until some remains unreacted, filter, partially evaporate the filtrate, cool it, then filter and dry the crystals.
- C** Add an indicator to the acid and titrate it with solid copper oxide.
- D** Dissolve the copper oxide in water, filter, then add sulfuric acid.
- E** Mix the reagents and collect the product by simple distillation.

WORKED SOLUTION

Copper(II) oxide is insoluble. Adding it in excess ensures that all the sulfuric acid has reacted:



Unreacted copper oxide is removed by filtration. The filtrate is then partially evaporated and cooled to crystallise copper(II) sulfate, after which the crystals are filtered and dried.

Therefore the answer is **B**.

QUESTION 16

Magnesium carbonate reacts with excess dilute hydrochloric acid.



Which measurement could be used to follow the rate of the reaction **without collecting the carbon dioxide gas**?

- A** Measure the total mass of the reaction vessel and contents at regular time intervals.
- B** Measure only the final pH.
- C** Measure the original volume of hydrochloric acid.
- D** Measure the mass of magnesium chloride before the reaction begins.
- E** Measure only the room temperature.

WORKED SOLUTION

Carbon dioxide leaves the reaction vessel as the reaction proceeds, so the total mass of the vessel and contents decreases.

Recording the mass at regular time intervals therefore provides a measure of the reaction progress and hence its rate.

The answer is **A**.

QUESTION 17

Three experiments use the same mass of calcium carbonate and excess hydrochloric acid at the same temperature.

experiment	calcium carbonate	acid concentration
1	chips	1.0 mol dm^{-3}
2	powder	1.0 mol dm^{-3}
3	chips	2.0 mol dm^{-3}

The same volume of acid is used each time.

Consider the statements.

1. Experiment 2 should reach completion sooner than experiment 1.
2. Experiment 3 should produce twice the final volume of carbon dioxide as experiment 1.
3. Experiments 2 and 3 should both have greater initial rates than experiment 1.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Powder has a larger surface area than chips, so experiment 2 has a greater rate and should reach completion sooner than experiment 1. Statement 1 is true.

Experiment 3 has a higher acid concentration, so it also has a greater initial rate. Statement 3 is true.

However, the same mass of calcium carbonate is used and the acid is in excess in every experiment. Calcium carbonate therefore determines the total amount of carbon dioxide, so the final gas volume is the same. Statement 2 is false.

Thus statements 1 and 3 only are correct: **F**.

C-18

Electrolysis and half-equations

Answer **B**

QUESTION 18

Molten magnesium bromide is electrolysed using inert electrodes.

Which row correctly explains why **direct current** is used and gives the half-equation at the **anode**?

option	reason for using dc	anode half-equation
A	keeps electrode polarity fixed	$\text{Mg}^{2+} + 2\text{e}^{-} \rightarrow \text{Mg}$
B	keeps electrode polarity fixed	$2\text{Br}^{-} \rightarrow \text{Br}_2 + 2\text{e}^{-}$
C	continually reverses electrode polarity	$2\text{Br}^{-} \rightarrow \text{Br}_2 + 2\text{e}^{-}$
D	continually reverses electrode polarity	$\text{Mg}^{2+} + 2\text{e}^{-} \rightarrow \text{Mg}$
E	prevents ions moving	$\text{Br}_2 + 2\text{e}^{-} \rightarrow 2\text{Br}^{-}$

A A

B B

C C

D D

E E

WORKED SOLUTION

Direct current keeps each electrode at a fixed polarity. With alternating current the electrode polarity would repeatedly reverse.

At the positive electrode, bromide ions are oxidised:



Therefore row **B** is correct.

C-19

Hydrocarbon properties

Answer **C**

QUESTION 19

As the carbon-chain length of straight-chain alkanes increases from C_4 to C_8 , which combination of trends is expected?

option	boiling point	viscosity	flammability
A	decreases	decreases	decreases
B	increases	decreases	increases
C	increases	increases	decreases
D	decreases	increases	decreases
E	increases	decreases	decreases

A A

B B

C C

D D

E E

WORKED SOLUTION

As chain length increases, intermolecular attractions become stronger.

Therefore boiling point and viscosity increase. Larger hydrocarbons are less volatile and generally less readily flammable.

The correct combination is **increase, increase, decrease**, so the answer is **C**.

C-20

Structural isomerism

Answer **B**

QUESTION 20

Which pair consists of structural isomers?

A CH_4 and C_2H_6

B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$

C $\text{CH}_2=\text{CH}_2$ and CH_3CH_3

D $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3COOH

E $\text{CH}_3\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

WORKED SOLUTION

Structural isomers have the same molecular formula but different structural formulae.

Both butane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$, and methylpropane, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$, have molecular formula C_4H_{10} , but the atoms are connected differently.

Therefore the answer is **B**.

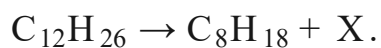
C-21

Cracking

Answer **C**

QUESTION 21

A hydrocarbon undergoes cracking:



What is X?

A C_2H_4

B C_3H_6

C C_4H_8

D C_4H_{10}

E C_5H_{10}

WORKED SOLUTION

Balance the carbon atoms:

$$12 = 8 + 4,$$

so X contains four carbon atoms.

Balance the hydrogen atoms:

$$26 = 18 + 8.$$

Therefore $X = C_4H_8$, which is consistent with cracking producing an alkene.

The answer is **C**.

C-22

Reactions of alkenes

Answer **D**

QUESTION 22

Ethene is shaken with bromine water.

Which row correctly gives the observation and the organic product?

option	observation	organic product
A	bromine colour remains	C_2H_6
B	bromine water decolourises	C_2H_6
C	bromine water decolourises	C_2H_5Br
D	bromine water decolourises	$C_2H_4Br_2$
E	solution turns blue	$C_2H_4Br_2$

A A

B B

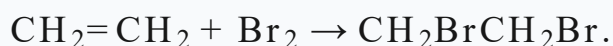
C C

D D

E E

WORKED SOLUTION

Bromine adds across the carbon-carbon double bond in ethene:



The bromine-water colour disappears and the product has molecular formula $C_2H_4Br_2$.

Therefore row **D** is correct.

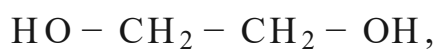
C-23

Condensation polymers

Answer **B**

QUESTION 23

A diol,



reacts with a dicarboxylic acid,



Which row correctly identifies the type of polymer formed and the small molecule eliminated during polymerisation?

- A** polyester; hydrogen
- B** polyester; water
- C** polyamide; water
- D** addition polymer; no small molecule
- E** polyamide; carbon dioxide

WORKED SOLUTION

A diol reacting with a dicarboxylic acid forms ester links, $-COO-$, between monomer units.

The polymer is therefore a polyester. Each condensation step eliminates a molecule of water.

The answer is **B**.

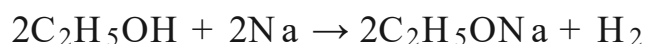
C-24

Alcohols

Answer **C**

QUESTION 24

Ethanol reacts with excess sodium.



What volume of hydrogen gas at rtp is formed when 4.60 g of ethanol reacts completely?

$$M_r(\text{C}_2\text{H}_5\text{OH}) = 46$$

and

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

A 0.60 dm³

B 0.80 dm³

C 1.20 dm³

D 2.40 dm³

E 4.80 dm³

WORKED SOLUTION

The amount of ethanol is

$$n = \frac{4.60}{46} = 0.100 \text{ mol.}$$

From the equation, two moles of ethanol produce one mole of hydrogen, so

$$n(\text{H}_2) = 0.0500 \text{ mol.}$$

At rtp,

$$V = 0.0500 \times 24.0 = 1.20 \text{ dm}^3.$$

The answer is **C**.

C-25

Carboxylic acids

Answer **C**

QUESTION 25

Ethanoic acid reacts completely with calcium carbonate.

Which pair gives the salt formed and the gas released?

A CaCO_3 ; H_2

B $\text{Ca}(\text{CH}_3\text{COO})$; CO_2

C $\text{Ca}(\text{CH}_3\text{COO})_2$; CO_2

D $\text{Ca}(\text{CH}_3\text{COO})_2$; H_2

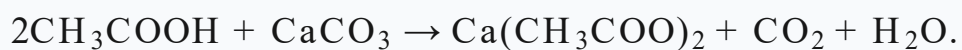
E CH_3COOCa ; O_2

WORKED SOLUTION

An acid reacting with a carbonate produces a salt, carbon dioxide and water.

The ethanoate ion is CH_3COO^- and calcium is Ca^{2+} , so the salt is $\text{Ca}(\text{CH}_3\text{COO})_2$.

The balanced reaction is



Therefore the answer is **C**.

QUESTION 26

Consider these statements.

1. A metal that is more reactive than carbon cannot normally be extracted from its oxide by reduction with carbon.
2. Compounds of transition metals are often coloured.
3. Transition metals form ions in only one oxidation state.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: metals above carbon in the reactivity series cannot normally be extracted from their oxides using carbon reduction.

Statement 2 is correct: coloured compounds are a common characteristic of transition metals.

Statement 3 is false: transition metals commonly form ions in more than one oxidation state.

Therefore statements 1 and 2 only are correct, so the answer is **E**.

C-27

Chemical tests

Answer **C**

QUESTION 27

An unknown aqueous salt gives:

- a **crimson-red flame**;
- a **white precipitate** when aqueous barium chloride is added in the presence of dilute hydrochloric acid.

Which salt is the most likely identity of the unknown?

A LiCl

B Na₂SO₄

C Li₂SO₄

D K₂SO₄

E CuSO₄

F LiNO₃

WORKED SOLUTION

A crimson-red flame identifies lithium ions, Li⁺.

A white precipitate with acidified barium chloride indicates sulfate ions, SO₄²⁻, because barium sulfate is insoluble and white.

The salt must therefore be Li₂SO₄.

The answer is **C**.

Compilation record: Level 0 uses Engineering Mock Test 1 for Mathematics/Physics and the corresponding Easy Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.